James Webb Space Telescope

Webb is the premier observatory of the next decade, serving thousands of astronomers worldwide. It studies every phase in the history of our Universe.

Webb studies every phase in the history of our Universe, ranging from the first luminous glows after the Big Bang, to the formation of solar systems capable of supporting life on planets like Earth, to the evolution of our own Solar System. Webb launched on Dec. 25th 2021. It does not orbit around the Earth like the Hubble Space Telescope, it orbits the Sun 1.5 million kilometers (1 million miles) away from the Earth at what is called the second Lagrange point or L2.

Key Facts

SPF 1 Million

Webb has a 5-layer sunshield that protects the telescope from the infrared radiation of the Sun, Earth, and Moon; like having sun protection of SPF 1 million.

Unprecedented Sensitivity

iIt will peer back in time over 13.5 billion years to see the first galaxies born after the Big Bang.in the ISS.

Premiere Telescope of Next Decade

extending the tantalizing discoveries of the Hubble Space Telescope.

Folding Design

So big it has to fold origami-style to fit in the rocket and will unfold like a “Transformer” in space.

1.5 Million km

Webb orbits the Sun 1.5 million kilometers from the Earth. (Hubble orbits 560 kilometers above the Earth.)

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Featured Image

The mid-infrared view of Centaurus A from NASA’s James Webb Space Telescope reveals dusty structures and hidden activity within the nearby, active galaxy.

NASA Webb Uncovers Unusual Galaxy Shaped by Cosmic Collision

In new images from NASA’s James Webb Space Telescope to celebrate its fourth science anniversary, a familiar galaxy transforms into...

Latest News

The feed below is a mix of Webb Science Releases and associated images/videos, Webb Science Blogs, Webb press releases and articles tagged with "James Webb Space Telescope". To filter out everything except the Webb Science Releases (or Science Blogs) - simply select that filter check box. Webb Science Releases, images, videos and Webb's Science Blogs can also be filtered by Webb's four primary science themes. Filter, Search and Sort Notes: Initial filtered results displayed below show articles that match all of the filters listed for all dates. Once one or more article type filters checked (Images, Science Blogs, Science Releases, Videos), only articles that match the checked filters (logical ORd) are displayed within the date range if selected. When a Webb-Theme is selected, that filters a subset of the article types selected. The "Search Latest Content" box searches for the terms entered within the subset of filtered articles. You can also sort the results by date in descending (newest first) or ascending (oldest first).

Filters

Helix Nebula Mini Poster

Download and print a mini poster featuring an image of the Helix Nebula, captured in 2024 by NASA's James Webb...

NASA Webb Uncovers Unusual Galaxy Shaped by Cosmic Collision

In new images from NASA’s James Webb Space Telescope to celebrate its fourth science anniversary, a familiar galaxy transforms into...

Centaurus A (MIRI Image)

The mid-infrared view of Centaurus A from NASA’s James Webb Space Telescope reveals dusty structures and hidden activity within the...

Centaurus A Crop (NIRCam and MIRI Image)

In the combined mid- and near-infrared view of Centaurus A, the NIRCam (Near-Infrared Camera) on NASA’s James Webb Space Telescope...

Centaurus A Context Image (ESO and Webb Images)

A ground-based image of nearby galaxy Centaurus A from the European Southern Observatory (top left) puts the near-infrared and mid-infrared...

Centaurus A (MIRI Compass Image)

Annotated image of the active galaxy Centaurus A captured by the James Webb Space Telescope's MIRI (Mid-Infrared Instrument), with compass...

Centaurus A Crop (NIRCam and MIRI Compass Image)

Annotated image of the active galaxy Centaurus A captured by the James Webb Space Telescope's NIRCam (Near-Infrared Camera) and MIRI...

NASA's Webb Reveals Stars Sparking to Life in Cosmic Celebration

NASA's James Webb Space Telescope has captured the infrared light of numerous features that previously were impossible to see beyond...

FS Tau (Webb Image)

In infrared light, NASA's James Webb Space Telescope reveals bright protostars in star system FS Tau and a tapestry of...

FS Tau Side-by-Side (Webb and Hubble Image)

A comparison between the observations of FS Tau by NASA's Hubble and James Webb space telescopes. Hubble's visible-light view shows...

FS Tau (Webb Compass Image)

An image of FS Tau captured by Webb's NIRCam (Near-Infrared Camera), with compass arrows, scale bar, and color key for...

NASA's Webb Studies How Planet Survived Death of its Star

NASA's James Webb Space Telescope is giving us new insight into the far-future of solar systems like our own, as...

Webb's Blog

This feed only contains articles from Webb's Blog which offers an insider's point of view covering a variety of topics. The feed below supports a deeper dive into Webb's Blog which contains early release science (not yet peer reviewed) as well as blogs on Webb Mission Operations and Webb Engineering posts - you can also filter Webb Science Blogs by Webb's four primary science themes.

Filters

NASA’s Webb Detects Methane on Interstellar Comet 3I/ATLAS

NASA’s James Webb Space Telescope has collected its first mid-infrared chemical fingerprint of an interstellar object during a recent revisit…

How NASA’s Webb Helped Rule Out Asteroid’s Chance of 2032 Lunar Impact

Editor’s Note: This post highlights data from Webb science in progress, which has not yet been through the peer-review process.…

New NASA Asteroid Observations Eliminate Chance of 2032 Lunar Impact

Using data from NASA’s James Webb Space Telescope observations collected on Feb. 18 and 26, experts from NASA’s Center for…

NASA’s Webb Space Telescope Observes Interstellar Comet

NASA’s James Webb Space Telescope observed interstellar comet 3I/ATLAS Aug. 6, with its Near-Infrared Spectrograph instrument. The research team has…

New Moon Discovered Orbiting Uranus Using NASA’s Webb Telescope

Editor’s Note: This post highlights data from Webb science in progress, which has not yet been through the peer-review process.…

NASA’s Webb Finds Possible ‘Direct Collapse’ Black Hole

Editor’s Note: This post highlights a combination of peer-reviewed results and data from Webb science in progress, which has not…

NASA’s Webb Observations Update Asteroid 2024 YR4’s Lunar Impact Odds

While asteroid 2024 YR4 is currently too distant to detect with telescopes from Earth, NASA’s James Webb Space Telescope collected one…

How NASA’s Webb Telescope Supports Our Search for Life Beyond Earth

This artist’s concept shows what exoplanet K2-18 b could look like based on science data. K2-18 b, an exoplanet 8.6…

NASA’s Webb Finds Asteroid 2024 YR4 Is Building-Sized

Editor’s Note: This post highlights data from Webb science in progress, which has not yet been through the peer-review process. These…

NASA’s Webb Reveals the Ancient Surfaces of Trans-Neptunian Objects

Trans-Neptunian objects (TNOs) are icy bodies ranging in size from Pluto and Eris (dwarf planets with diameters of about 1,500…

Monitoring Webb’s Mirrors for Optimal Optics

NASA’s James Webb Space Telescope is the largest and most powerful telescope ever launched to space. Its mirror is composed…

Webb Researchers Discover Lensed Supernova, Confirm Hubble Tension

Editor’s Note: This post highlights data from Webb science in progress, which has not yet been through the peer-review process. Measuring…

Latest 2026 Images

The image below is a SLIDESHOW. Hover over the image to see the image title and controls. Click the image to go to a detail page with more info and the ability to download the image at various resolutions (click the downward arrow icon in lower right corner).

Science Images By Year

Webb Science image slideshows by year.

First Images

Webb's first science images released in July of 2022 just after commissioning was completed.

More Webb Images

Images from Webb's testing and commissioning, road to launch from the US to ESA's launch facility in French Guiana, Webb's launch, Webb's development and more.

About Webb Images

The where, why and how of Webb images.

Search Webb Science Images

Search and filter Webb science images and image products using keywords, date ranges, astronomoical objects and more.

What is Webb Observing?

See current, upcoming and recent past observations scientists are making with the Webb Space Telescope. View details about each observation's science focus areas, the instruments used and more.

The Webb Mission

Webb is the premier observatory of the next decade, serving thousands of astronomers worldwide. It studies every phase in the history of our Universe, ranging from the first luminous glows after the Big Bang, to the formation of solar systems capable of supporting life on planets like Earth, to the evolution of our own Solar System.

This is a photo of one of the James Webb Space Telescope's primary mirror segments coated with gold by Quantum Coating Incorporated. It's NOT a flight segment, it's the engineering design unit. The photo was taken at BATC by Drew Noel.

Webb's Science Goals

The James Webb Space Telescope is a giant leap forward in our quest to understand the Universe and our origins. Webb is examining every phase of cosmic history: from the first luminous glows after the Big Bang to the formation of galaxies, stars, and planets to the evolution of our own solar system. Learn about the 4 main science themes for Webb.

NASA’s Webb Captures Dying Star’s Final ‘Performance’ in Fine Detail

The Spacecraft

The Webb Space Telescope is the largest, most powerful and most complex telescope ever launched into space . It's design and development history stretches back before the Hubble Space Telescope was launched. Learn about the design, the major components and subsystems of Webb and see Webb in 3d in a 3d Solar System.

The International Webb Team

Webb is for the world, and from the world. Thousands of skilled scientists, engineers and technicians from 14 countries (and more than 29 U.S. states, and Washington, D.C.)

contributed to the design, build, test, integration, launch, commissioning and operations of Webb. It is a joint NASA/ESA/CSA mission. Assembly and testing of the mirror and instruments occurred at NASA Goddard (GSFC).

A full disk view of the earth from GOES 16, GOES East on the vernal Equinox.

Do NASA Science with Galaxy Zoo!

Join NASA researchers and help discover the secrets of the universe.

NASA invites people of all ages and backgrounds to participate in authentic NASA research via "citizen science" or "participatory science" projects, where volunteers and amateurs have helped make thousands of important scientific discoveries! As a part of the Galaxy Zoo project, you can help classify the latest images of galaxies from NASA's Webb telescope.

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